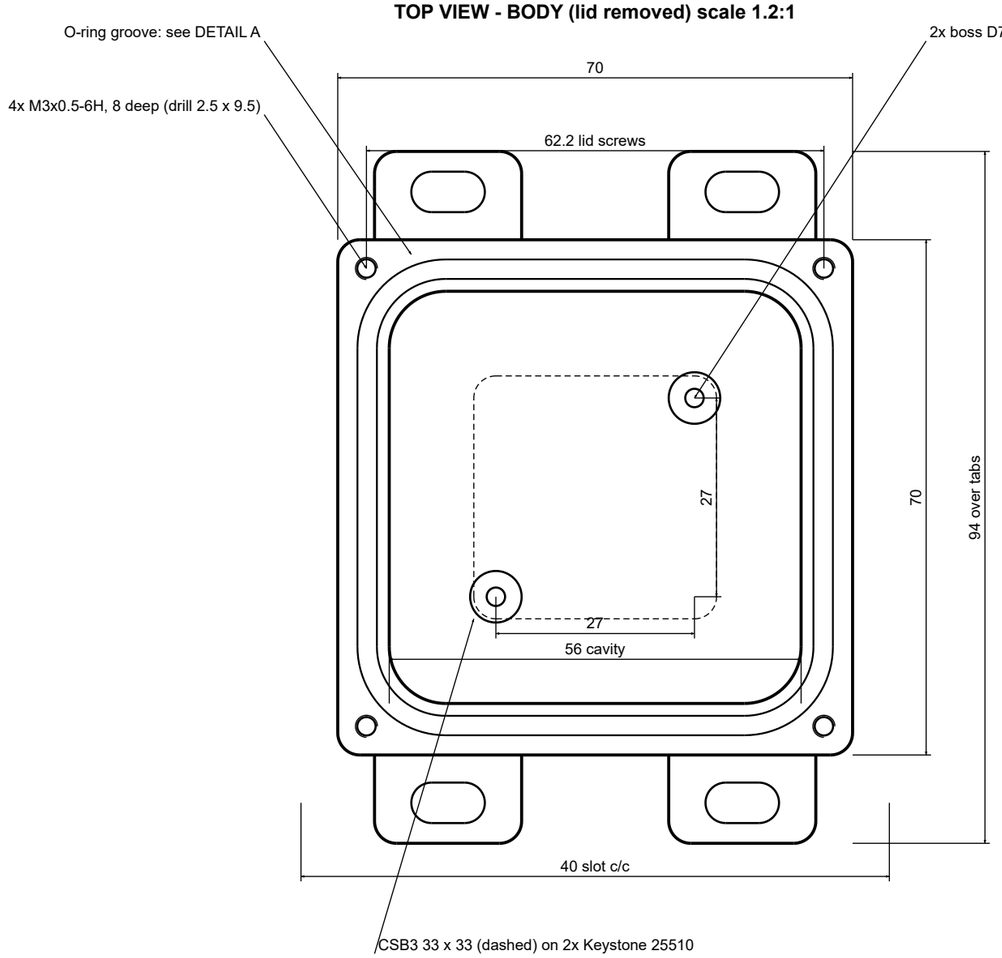
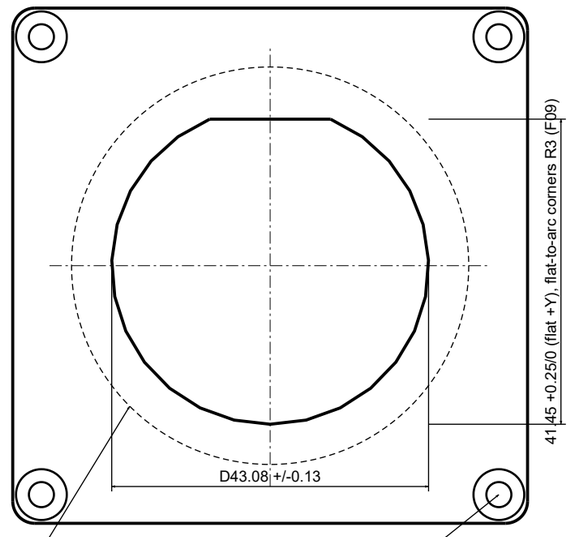
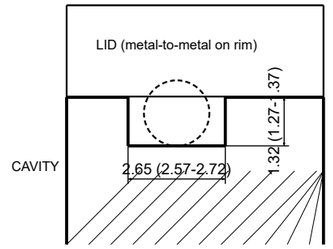
TOP VIEW - BODY (lid removed) scale 1.2:1



LID 70 x 70 x 6 (top) scale 1.2:1



DETAIL A - O-ring groove scale 6:1

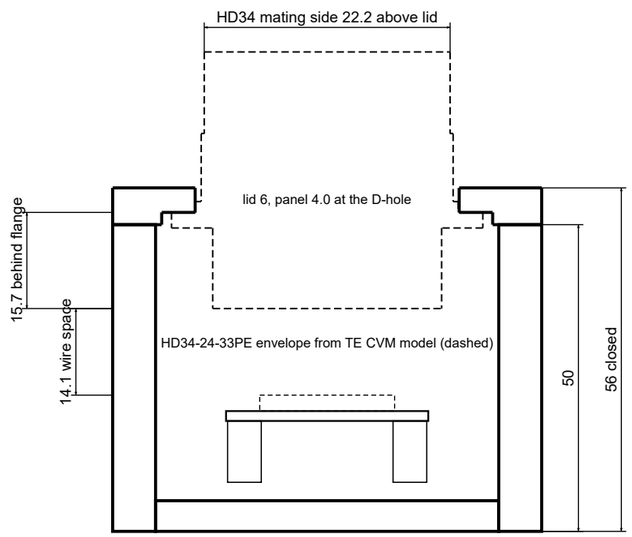


Parker ORD 5700 Design Chart 4-3, W 1.78 face seal, liquids

groove R 0.13-0.38, break edges 0.13, Ra 0.8 (32 rms) groove + lid face

squeeze 26 % nominal, fill 71 %; AS568-039 stretched ~2.4 % on centreline

SECTION B-B (through connector) scale 1:1



NOTES - CSB3 box

1. 6061-T6 billet body + lid. Stock est.: body 75 x 100 x 53, lid 75 x 75 x 8.
2. Seal: AS568-039 (same groove length 229.6 as the VRC box), compound TBD (E4).
3. Lid: 4x M3 x 12 ISO 10642 A2 countersunk, tapped holes outside the O-ring.
4. Connector on LID: TE Deutsch HD34-24-33PE, jam nut. Cutout per TE 114-151014
rev A 3.3-I (D43.08 / 41.45 flat, seal face D50.80, panel 1.59-4.76).
Flange inside on the D54 seat; lockwasher 112264 + panel nut 112263-90 outside,
7.4-8.4 N.m (114-151014 3.11). Nut D47.63 needs a 1-7/8 in socket above the lid.
Geometry behind the flange (15.7) read from TE CVM model c-hd34-24-33pe-a-3d.stp.
5. CSB3 (33 x 33 x 4.21, 2x D3.5 holes; ECUMaster manual v2.1 p.4) on 2x Keystone
25510 nylon standoffs (2x M3 x 8 A2 set screw boss -> standoff, F03). 14.1 mm board parts to HD34 rear.
6. Mating HD36-24-33SE plug + boot sits above the lid: keep about 70 mm clear.
7. Mount on 4x M6 nylon screws (E5), slots 6.6 x 10 (F01). BREAK SHARP EDGES 0.2-0.5 EXCEPT O-RING / SEAL FACES.
8. 120 ohm terminator jumper stays OPEN (csb3-bom.csv).
9. Tab corners R3 at x +-10; R1.9 at x +-30 (2.0 ledge to the body corner): CUTTER r <= 1.5 there (F11).
10. Floor-to-wall inside corners: CUTTER CORNER OK. CSB3 board in CAD is an envelope, not ECUMaster CAD (F17).

ST185 CSB3 ENCLOSURE

Body + lid, O-ring face seal, HD34 jam-nut receptacle on the lid

Units mm | 3rd-angle | scales as noted | 6061-T6 | General tol. ISO 2768-m unless noted

Claude Code 2026-10-13, rev 2026-10-16 (SteveCAD audit v3) | Fusion ST185-CSB3-Enclosure v3 | DRAFT for Dan review